Stacking and Blending Ensembles

These are advanced ensemble learning techniques used to combine multiple base machine learning models via a "meta-learner" to improve predictive performance.

- Stacking uses K-Fold cross validation to generate training data for the meta-learner.
- Blending uses a simpler holdout validation set
- These methods reduce error by leveraging diverse model strengths.

* Stacking (Stacked Generalization)

Process -

Base models are trained on the full dataset, but their predictions are generated using K-Fold cross-validation to create "out of Fold" predictions.

Data Usage -

It uses all available training data to train the meta-learner, making it more efficient and less prone to data leakage.

Complexity

Generally more computationally expensive due to multiple training passes.

* Blending

Process : Trains base models on a training set and makes predictions on a separate "holdout" validation set.

meta-model

The meta-learner is trained exclusively on the predictions made on this holdout set.

Complexity

Generally faster and simpler to implement than stacking.

When to use

stacking is often preferred in competitions for max accuracy.

Blending is preferred when computational resources are limited or for faster devemnt